

Out of the Darkness and into the Light? Development Effects of Rural Electrification

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Nearly 1 billion people still lack electricity access. Developing countries are investing billions of dollars in “last-mile” electrification, although evidence on its economic impacts is mixed. We estimate the development effects of rural electrification in the context of India’s national electrification program, RGGVY (Rajiv Gandhi Grameen Vidyutikaran Yojana), which reached over 400,000 villages. Using regression-discontinuity and difference-in-differences designs, we estimate that RGGVY meaningfully expanded electricity access. However, the program generated limited economic impacts after 3–5 years. Scaling our intent-to-treat estimates using instrumental variables, we find that “full electrification” reduces welfare in small villages but has a 33% internal rate of return in large villages.

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I. Introduction

Nearly 1 billion people still lack access to electricity, despite substantial investments to extend the power grid across the developing world.¹ The International Energy Agency projects that achieving universal electrification will cost \$49 billion per year between 2019 and 2030. The vast majority of remaining unconnected households live in rural South Asia and sub-Saharan Africa (IEA 2019).

While electricity access is highly correlated with GDP (gross domestic product) at the national level, prior research on the causal effects of electrification has produced mixed results. Seminal early work finds large positive impacts of electrification on development outcomes (Dinkelman 2011; Rud 2012; Lipscomb, Mobarak, and Barham 2013). In contrast, recent experimental evidence finds rural electrification to be welfare reducing, with negligible benefits and large costs (Lee, Miguel, and Wolfram 2020b), and estimates only modest welfare gains from expanding access to grid power (Burgess et al. 2023). Such discrepancies may reflect differences in scale: while estimates of large economic impacts have tended to come from electrifying large populations (e.g., entire Indian states in Rud 2012), studies finding less favorable welfare impacts have been conducted at the village level—a scale that is more representative of today's electrification efforts.

In this paper, we estimate the economic impacts of electrification in the context of India's massive national rural electrification program, Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY). The "Prime Minister's Rural Electrification Program" was launched in 2005 to expand both domestic and commercial electricity access in over 400,000 rural villages across 27 Indian states. India is a useful setting for studying ongoing rural electrification efforts, as it contributed over 80% of global gains in new household grid connections between 2000 and 2016 (IEA 2017). Moreover, India's per capita income during the RGGVY period was similar to income levels in countries with significant unelectrified populations today (see fig. 1). The program's scope also provides a unique opportunity to address the divergent results from the existing literature.

We use two key features of RGGVY's implementation to estimate the program's impacts on both electricity access and economic outcomes. First, villages were eligible for electrification under RGGVY only if they

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¹ Roughly 800 million people remained unelectrified in 2018, down from 1.2 billion in 2010 (Tracking SDG7 2020). Universal energy access is United Nations Sustainable Development Goal 7 (UNDP 2015), and electrification is a key piece of the World Bank's investment strategy (World Bank 2015).

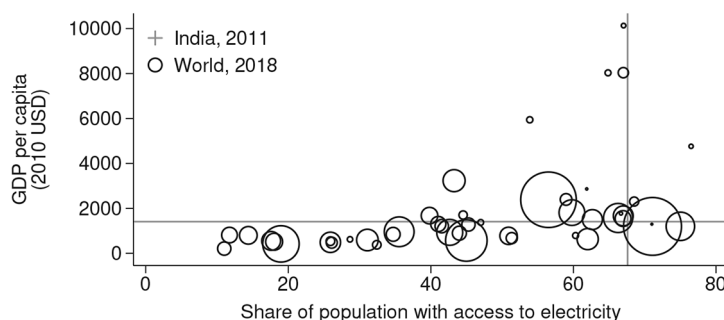


FIG. 1.—Electricity access and per capita GDP: India versus the world. This figure plots country-level per capita GDP (in 2010 USD) versus the share of the population with electricity access. Gray lines indicate India's levels in 2011, at the end of our study period. Circles show all 46 countries with below 80% electricity access in 2018, with a total population of 1.28 billion people. Circle sizes are scaled by each country's population. Twenty-eight countries (containing 66% of the people) had 2018 per capita GDPs lower than India's 2011 per capita GDP. An additional five countries (containing 11% of the people) had per capita GDPs within 20% of India's 2011 level. Data are from World Bank (2021).

contained at least one neighborhood (“habitation”) larger than 300 people.² This allows us to estimate a regression-discontinuity (RD) design, using this population-based eligibility threshold. Second, RGGVY had a staggered rollout, treating districts in two waves corresponding to India's 10th and 11th Five-Year Plans. This facilitates a difference-in-differences (DD) design comparing first- and second-wave districts. Using both administrative and geospatial data, we apply our RD strategy to a sample of over 10,000 villages with close to 300 people and our DD strategy to nearly all of rural India.

We first show that RGGVY led to substantial increases in electricity access. We find that RGGVY provided commercial power to 1 in 13 barely eligible 300-person villages that previously lacked access, while increasing average commercial power supply by 0.56 hours per day.³ We also find that RGGVY electrified one in seven previously unconnected rural households in first-wave districts and increased average household electricity consumption by 4 kilowatt-hours (kWh) per month. Consistent with these direct measures of electrification, we estimate 4–5 percentage point (pp) increases in household electric lighting adoption. We detect corresponding

² The village was the lowest-level administrative unit in the 2001 census of India. Villages are composed of “habitations” (or “hamlets”), which correspond to distinct inhabited areas within a village. South Asian villages typically have one or more inhabited regions surrounded by agricultural land. India's 600,000 villages contain approximately 1.6 million unique habitations.

³ These two estimates are internally consistent, as the average electrified village in this setting received under 11 hours per day of commercial power supply.

increases in satellite-derived nighttime brightness at the village level, using both our cross-sectional RD design and a DD event-study model. These results tell a consistent story: while RGGVY fell short of achieving “full electrification,” it succeeded in meaningfully expanding electricity access and consumption in rural India.⁴ However, despite these gains in electrification, we find that RGGVY led to at most modest changes in economic outcomes. We can reject intent-to-treat effects on our preferred outcome, per capita consumption expenditure, greater than 2% of the mean using our RD strategy and 8% of the mean using our DD approach.

Next, we rescale our estimates of RGGVY’s impacts via instrumental variables (IVs), in order to estimate the effects of rural electrification more broadly. Using a fuzzy RD design, we estimate statistically insignificant decreases in per capita expenditure as a result of “full electrification” in villages with close to 300 people. Using a DD-IV design, we also find statistically insignificant decreases in per capita expenditure from “full electrification” for smaller villages (median of 1,043 people). We can reject expenditure increases greater than 26% (fuzzy RD) and 30% (DD-IV) in smaller villages. On the other hand, in larger villages (median of 2,076 people), our DD-IV design cannot reject a tripling of per capita expenditure due to “full electrification.” We see suggestive evidence that these results may be driven by structural transformation: we find evidence of firm growth only in larger villages.

Finally, we use our estimates to conduct a welfare analysis. We compute the 20-year returns from rural electrification using two strategies for quantifying benefits: (i) our fuzzy RD and DD-IV per capita expenditure estimates and (ii) consumer surplus from household electricity consumption.⁵ Both strategies imply that “full electrification” is benefit-cost negative in small villages and benefit-cost positive in large villages. For 300-person villages, we calculate a 0% internal rate of return (IRR), which is welfare reducing under any time discounting. For 1,000-person villages, our 13% IRR just exceeds a standard 10%–12% benchmark for cost-effectiveness. For 2,000-person villages, our 33% IRR far exceeds this benchmark, reflecting both greater per capita benefits in larger villages and economies of scale in costs. These results help to reconcile divergent estimates from the literature—electrification reduces welfare in small villages and increases welfare in larger communities.

This paper makes three key contributions. First, we provide new evidence on the economic impacts of rural electrification from across India—home to the world’s largest unelectrified population during our

⁴ These findings align with Indian policy reports on RGGVY (e.g., Sreekumar and Dixit 2011; PEO 2014; Josey and Sreekumar 2015).

⁵ Throughout this paper, we use the term “welfare” to refer to the benefits of electrification minus its costs. If a reader prefers not to take a stand on a particular social welfare function, they may interpret our results in terms of the “social surplus” of electrification.

sample period. Our estimates leverage policy variation from a flagship electrification program; they come from thousands of villages and hundreds of districts, in a setting with income levels comparable to those on today's electrification frontier.

Second, we add to the knowledge on the causal effects of infrastructure in developing countries. Whereas existing research has tended to find large positive impacts of infrastructure investments, our results indicate that "last-mile" infrastructure projects may be less likely to spur economic growth.⁶ Third, we contribute to a growing literature on electricity in the developing world, with our finding of heterogeneous impacts of multi-sector (residential and nonresidential) electrification.⁷ Our results speak to two hypotheses proposed by Lee, Miguel, and Wolfram (2020a): (i) that the benefits of electrification vary across local economic settings, which we characterize in terms of heterogeneous village size, and (ii) that household electrification alone may be insufficient for economic gains. Even though RGGVY expanded access beyond the residential sector, we find that electrifying small villages reduces welfare. We find positive welfare impacts in larger villages, accompanied by suggestive evidence of structural transformation.

II. The RGGVY Electrification Program

Upon its independence in 1947, only 1,500 of India's villages had access to electricity (Tsujita 2014). By 2004, after decades of electrification efforts, over 125,000 rural villages still lacked power access. In the remaining 467,000 villages, electrification was often extremely limited, with 57% of all rural households lacking grid connections.⁸ In 2005, the national government launched the flagship RGGVY program, which sought to

⁶ For example, Donaldson (2018) finds that early railroad investments increased real incomes in India, and Faber (2014) finds that highway infrastructure investments led to potentially large aggregate efficiency improvements in China. Conversely, Asher and Novosad (2020) do not detect meaningful economic impacts from PMGSY (Pradhan Mantri Gram Sadak Yojana), India's "last-mile" rural road construction program.

⁷ Several papers highlight heterogeneity at the intersection of electricity and economic development. For example, Gertler et al. (2016) find that income levels are a key driver of energy demand in Mexico; Allcott, Collard-Wexler, and O'Connell (2016) shows that generator ownership determines firms' responses to power outages in India; and Ryan (2020) and Mahadevan (forthcoming) present evidence on disparate treatment of politically connected firms and individuals in energy contracts and power supply, respectively.

⁸ Grid power is not the only source of electricity in rural villages. Data from a 2015 household survey report off-grid electricity access from microgrids, solar home systems, and diesel generators (Aklin et al. 2016). In 2013, households in Bihar could access off-grid electricity from diesel generators, solar microgrids, or their own solar systems (Burgess et al. 2023). However, off-grid solar access was extremely limited before 2012 (Lighting Global and GOGLA 2017), leaving diesel generators as the main alternative power source during our sample period.

(i) connect over 100,000 unelectrified rural villages and (ii) more intensively electrify over 300,000 “underelectrified” villages.⁹

RGGVY had a dual mandate to install electricity infrastructure to support village economies and connect unelectrified households. Infrastructure investments—transmission lines, distribution lines, and transformers—aimed to “facilitate overall rural development, employment generation, and poverty alleviation” by supporting electric irrigation pumps, microenterprises, and small-to-medium industries (Ministry of Power 2005). New infrastructure also extended the grid to public places such as schools, health clinics, and local government offices. To increase residential power access, RGGVY was charged with providing free grid connections to below-poverty-line households.¹⁰ RGGVY targeted both the extensive and intensive margins, connecting new villages to the grid while also upgrading existing infrastructure and connecting additional households in villages with some degree of electrification before 2005.

In order to receive RGGVY funding, states submitted Detailed Project Reports (DPRs) to the central government, based on village-level surveys conducted by local electricity utilities. Each DPR proposed a village-by-village implementation plan for a particular district, including specific infrastructure to be installed and the number of households to be connected. The Rural Electrification Corporation (overseen by the Ministry of Power) reviewed DPRs, approved projects, and disbursed funds to states. By 2011, RGGVY had provided over Rs 253 billion (US\$5.45 billion) in funds and connected 17.5 million households to the grid—roughly one in five previously unelectrified rural households in India (Sreekumar and Dixit 2011). RGGVY’s setting is also instructive about ongoing electrification efforts: figure 1 shows that India’s 2011 GDP per capita is comparable to the 2018 GDP per capita in countries where a substantial share of the population remains unelectrified today.

Two features of RGGVY’s implementation facilitate our empirical analysis. First, DPRs were funded under India’s Five-Year Plans, and districts were sorted first-come-first-served into two waves. The first wave (229 districts) were authorized under the 10th Plan and received funding between 2005 and 2008. The second group (331 districts) were authorized under the 11th Plan and received funds between 2008 and 2011. Approximately 164,000 (267,000) villages and 7.5 million (14.6 million) below-poverty-line households were slated for electrification under the 10th (11th) Plan. Figure 2 maps districts according to their Five-Year Plan; 23 of

⁹ RGGVY translates to “The Prime Minister’s Rural Electrification Plan.” It was subsequently subsumed into Deendayal Upadhyaya Gram Jyoti Yojana (DDUGJY), the scheme launched in 2015 with the goal of providing continuous 24×7 power to all of rural India.

¹⁰ Above-poverty-line households were able to purchase connections. RGGVY did not alter electricity prices, but Indian retail electricity tariffs are heavily subsidized (Burgess et al. 2020). In 2010, the median rural tariff was 2.64 rupees (Rs; US\$0.05) per kilowatt-hour.

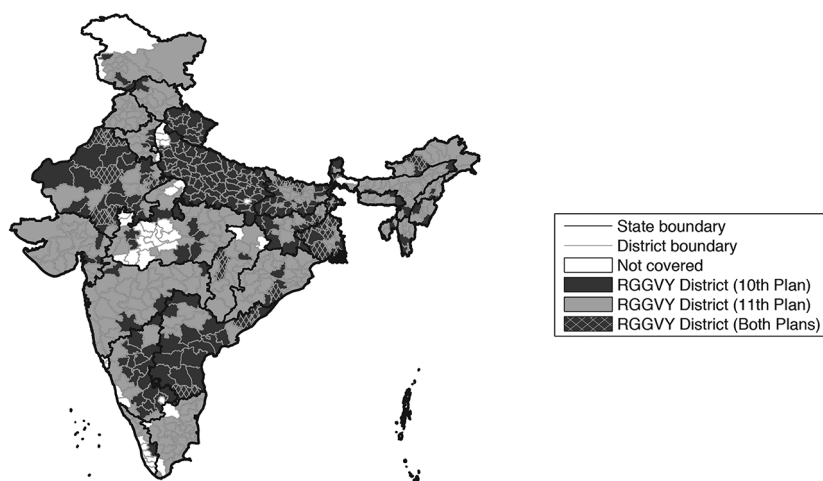


FIG. 2.—Indian districts by RGGVY implementation phase. We shade 2001 districts by RGGVY coverage status. Dark districts were covered under the 10th Plan (RGGVY's first wave), light districts were covered under the 11th Plan (RGGVY's second wave), cross-hatched districts were covered under both the 10th and 11th Plans, and white districts were not covered by RGGVY. In 2001, India had 584 districts across its 28 states and seven union territories. RGGVY covered 530 total districts in 27 states (neither Goa nor the union territories were eligible); 30 districts were split between the 10th and 11th Plans; 23 states contain both 10th- and 11th-Plan districts.

27 states contain both 10th- and 11th-Plan districts. We leverage this staggered rollout to estimate the impacts of electrification, comparing 10th- and 11th-Plan districts in a DD design.

Second, RGGVY determined village eligibility using the populations of subvillage “habitations” (i.e., neighborhoods). Under the 10th Plan, only villages with constituent habitations larger than 300 people were eligible for electrification. We use an RD design to identify local average treatment effects (LATEs) of electrification at this 300-person cutoff, comparing barely ineligible to barely eligible villages in 10th-Plan districts.¹¹ The RD and DD designs are complementary: our RD analysis uses weaker identifying assumptions and village-level variation but estimates effects local to the 300-person cutoff; our DD analysis requires stronger identifying assumptions using district-level variation but can estimate effects for villages of all sizes, inclusive of within-district spillovers.

¹¹ Under the 11th Plan, this threshold was decreased to 100 people. Because of the limited density of sub-100-person villages, we restrict our RD analysis to the 300-person cutoff in 10th-Plan districts. Focusing on RGGVY's earlier wave also lets us estimate economic impacts over a longer timescale of 3–5 years.

III. Empirical Strategy

A. RD Design

We estimate a sharp RD design using RGGVY's 300-person cutoff, where a village's eligibility for treatment switches from 0 to 1 as the running variable crosses 300. This running variable is technically the habitation population, since the 300-person cutoff applied at the subvillage level. Because data on subvillage population counts are unavailable, we restrict our RD sample to villages comprising exactly one habitation. For this 50% subset of villages, the village population is equivalent to the habitation population. This lets us use village population as a running variable to estimate the effects of RGGVY eligibility for single-habitation villages in 10th-Plan districts with populations close to 300.¹²

Our RD design necessitates two key identifying assumptions. First, we must assume continuity across the RD threshold for all village covariates and unobservables that might be correlated with our outcome variables. While this assumption is fundamentally untestable, pre-RGGVY village-level covariates appear to be smooth across the 300-person cutoff (see app. B.3; apps. A–C are available online). We are also unaware of any other Indian social program that uses 300 people as a salient criterion.¹³ Second, we assume that our running variable is not manipulable around the threshold. This assumption almost certainly holds, as program eligibility was contingent on populations from the 2001 census, enumerated 4 years before RGGVY's announcement in 2005. Figure 3 shows no evidence of bunching at the 300-person cutoff.

Under these assumptions, the following RD specification estimates the causal impact of RGGVY eligibility for villages with close to 300 people:

$$Y_v = \beta_0 + \beta_1 Z_v + \beta_2 (P_v - 300) + \beta_3 (P_v - 300) \cdot Z_v + \theta \mathbf{X}_v + \eta_s + \varepsilon_v \quad (1)$$

for $P_v \in [300 - h, 300 + h]$, where $Z_v \equiv \mathbf{1}[P_v \geq 300]$,

Y_v represents the outcome of interest in village v , P_v is the 2001 village population (the RD running variable), and Z_v is an indicator denoting RGGVY eligibility. To increase precision, we control for pre-RGGVY village-level covariates in $\mathbf{X}_v$, including the lagged outcome variable (where possible). We also include state fixed effects η_s . We implement equation (1) using the *rdrobust* framework developed by Calonico, Cattaneo,

¹² Our RD design is not suited for multihabitation villages, where the village population does not correspond to the habitation populations used to determine RGGVY eligibility. Panel A of table 1 shows that, within the neighborhood of the RD population threshold, single-habitation villages are similar to other villages in 10th-Plan districts on average.

¹³ Other Indian programs use population-based eligibility thresholds, including the PMGSY road construction program studied in Asher and Novosad (2020). PMGSY used 1,000- and 500-person cutoffs.

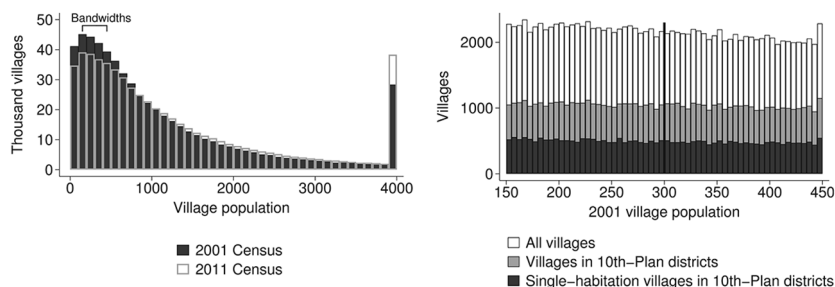


FIG. 3.—Density of RD running variable. The left-hand histogram shows village populations for 2001 (dark) and 2011 (light), top-coding each distribution at 4,000. The right-hand histogram zooms in on villages close to RGGVY's 300-person population cutoff, with 2001 populations between 150 and 450 (slightly wider than our optimal RD bandwidths). Dark bars show the sample of single-habitation 10th-Plan villages used in our RD analysis, relative to all Indian villages (white) and all villages in 10th-Plan districts (light gray).

and Titiunik (2014). Following the standard *rdrobust* procedures, we apply a triangular weighting kernel in distance from the RD cutoff, calculate MSE (mean squared error)-optimal RD bandwidths h , and use heteroskedasticity-robust nearest-neighbor standard errors.¹⁴

This sharp RD design estimates the impacts of eligibility for the RGGVY program local to the 300-person cutoff. We use an analogous fuzzy RD design to estimate the impacts of electrification, instrumenting for village-level electricity access using the 300-person eligibility indicator. This transforms our intent-to-treat RD estimates into LATEs that enable us to approximate a “full-electrification” program.¹⁵

B. DD Design

We complement our RD design with a DD analysis that leverages RGGVY's phased rollout. Our DD compares “treated” 10th-Plan districts (RGGVY's first wave) to a “control group” comprising both 11th-Plan districts (RGGVY's second wave) and non-RGGVY districts.¹⁶ Using district-level variation, this DD design lets us study villages of all sizes, incorporate district-level

¹⁴ The MSE-optimal bandwidth procedure computes a separate h for each outcome variable Y_i , following Calonico, Cattaneo, and Farrell (2018). We present *rdrobust* sensitivity analysis for alternative kernels, bandwidths, controls, functional forms, and standard errors in figs. B1 and B2 and tables B1–B3 (figs. A1–C6 and tables A1–C5 are available online).

¹⁵ Scaling up our intent-to-treat RD estimates also accounts for potential noncompliance with the 300-person eligibility rule. We implement fuzzy RD using the same *rdrobust* framework.

¹⁶ Thirty-seven districts in our control group had no RGGVY projects. We assign the 30 districts with RGGVY projects in both plans to the “treated” group. Our DD results are robust to dropping all 67 of these districts (see table B9).

outcome data, and capture within-district spillovers and general equilibrium effects.

Our main DD specification is

$$Y_{dt} = \gamma \mathbf{1}[\text{10th Plan}]_d \times \mathbf{1}[\text{Post-2005}]_t + \eta_d + \delta_t + \theta_{dt} + \varepsilon_{dt}, \quad (2)$$

where Y_{dt} is an outcome variable for district d in year t . The coefficient γ captures the differential impact of RGGVY eligibility after 10th-Plan districts began receiving treatment in 2005, controlling for district fixed effects η_d and year fixed effects δ_t . To account for potential selection of districts into RGGVY's first wave, θ_{dt} includes three sets of linear trends: (i) trends grouping districts within each state by quartiles of 2005 household expenditures, in case states prioritized electrifying poorer districts; (ii) trends grouping districts by national deciles of 2005 household expenditures, in case such selection existed in absolute terms across states; and (iii) state-specific trends. Our DD identifying assumption is that, after these trends are controlled for, 10th- and 11th-Plan districts would have continued on parallel counterfactual trajectories absent RGGVY. We fail to reject parallel trends before RGGVY.¹⁷

Equation (2) estimates the impacts of the RGGVY program over the full support of village populations. This DD approach complements our RD design, which estimates effects for 300-person villages. To estimate the impacts of electrification, we use the analogous DD-IV model, instrumenting for electricity access via the $\mathbf{1}[\text{10th Plan}]_d \times \mathbf{1}[\text{Post-2005}]_t$ interaction. This scales our intent-to-treat effects to account for the fact that RGGVY did not bring new electricity connections to every household. Since RGGVY prohibited hiring local workers, and since our end-line data were collected several years after most 10th-Plan projects, we are confident in the exclusion restriction that RGGVY eligibility affected economic outcomes only through the program itself.¹⁸

IV. Data

We use village-level data from the 2001 and 2011 censuses of India. The 2001 village population serves as our RD running variable, and we isolate the subset of single-habitation villages by matching villages to a separate

¹⁷ See table 1, fig. 6, and tables B7 and B8. After including state-specific trends, our pretrend estimates are all statistically indistinguishable from zero, except for TV ownership (which, if anything, suggests that 10th-Plan districts were trending away from TV ownership relative to 11th-Plan districts). For outcome variables with village-level granularity or annual frequency, we modify eq. (2) to include village fixed effects or multiple event-study γ coefficients.

¹⁸ Any exclusion restriction violation would involve a time-varying unobservable correlated with both economic outcomes and our instrument but not captured by household electricity access.

census of habitations.¹⁹ We observe village-level electricity access in the 2011 census, which reports (i) dummies for the village's first grid connection in the domestic, agricultural, and commercial sectors; (ii) average hours per day of grid power supplied to each sector; and (iii) average hours per day of grid power supplied to all three sectors. The census also provides a range of village-level economic outcome variables, including demographics, employment by gender and sector, household characteristics, asset ownership, and community-wide amenities.

We also incorporate data from a low-income subset of the 2011 Socioeconomic and Case Census (SECC). We observe income and wealth variables for this subset of households, and we also use the subset to reconstruct the share of poor households in each village.²⁰ Using the full (unrestricted) 2011 SECC, the Socioeconomic High-resolution Rural-Urban Geographic Dataset for India (SHRUG) estimates consumption expenditure per capita at the village level (Asher and Novosad 2020; Asher et al. 2021).²¹ We use SHRUG expenditure per capita as our preferred village-level outcome variable for quantifying the impacts of electrification on economic well-being.

We use administrative microdata from two additional sources. The Economic Census surveys all nonfarm establishments, which we use to construct counts of firms and firm employees in each village in 1990, 1998, 2005, and 2013.²² The District Information System on Education (DISE) provides student enrollment and pass rates for all Indian primary and upper primary schools from 2005 to 2014.²³ Panel A of table 1 reports pre-RGGVY summary statistics at the village level; our RD sample of single-habitation 10th-Plan villages appears quite similar to the universe of villages with populations between 150 and 450.

While census variables capture the extensive margin of village electricity access, we use satellite images of nighttime brightness to capture intensive-margin gains in electrification. The National Oceanic and Atmospheric Administration (NOAA) publishes annual composite images that report

¹⁹ Official RGGVY ledgers we observed in Rajasthan were preprinted with 2001 census populations. The National Rural Drinking Water Programme conducted habitation censuses in 2003 and 2009. We link these data to census villages by modifying a fuzzy matching algorithm from Asher and Novosad (2020). Appendix C.5 describes this matching algorithm; app. C.4 discusses the census data in further detail.

²⁰ Though the SECC enumerated the full population, we observe only households that met at least one of seven poverty indicators and none of 14 affluence indicators. See app. C.7 for further details.

²¹ SHRUG constructs village-level expenditure by combining 2011 SECC microdata with data from the Indian Human Development Survey (2011–12). To our knowledge, this is the only dataset of per capita expenditure that covers all Indian villages. See app. C.6 for further details.

²² This includes informal firms and public sector employers, as we discuss in app. C.8.

²³ DISE data include 1.68 million unique schools, and previous research has used them to measure student achievement (Adukia, Asher, and Novosad 2020). Appendix C.9 describes these data in detail.

TABLE 1
SUMMARY STATISTICS BEFORE RGGVY

	(1)	(2)	(3)
A. Village-Level Covariates, Population 150–450			
	All Districts	10th-Plan Districts	RD Sample
Agricultural workers/population (2001)	.39 (.16)	.37 (.16)	.40 (.15)
Nonagricultural workers/population (2001)	.06 (.08)	.06 (.08)	.06 (.08)
Number of firms in village (2005)	7.32 (11.30)	6.81 (10.78)	7.60 (12.23)
Literacy rate (2001)	.45 (.18)	.44 (.17)	.45 (.17)
School enrollment (2005–6 head count)	87.71 (188.18)	91.61 (140.95)	71.25 (115.25)
Electric access anywhere in village (2001)	.68 (.46)	.62 (.49)	.67 (.47)
Distance to nearest town (km)	27.72 (27.67)	24.69 (25.98)	23.81 (22.99)
Number of villages	129,438	62,638	18,686
B. District-Level Covariates, 2005 NSS			
	10th-Plan Districts	11th-Plan Districts	Pretrend Estimates
Expenditure per capita (Rs/month)	869.87 (231.73)	988.00 (371.21)	14.577 [20.719]
Share of HHs consuming any electricity	.46 (.31)	.66 (.28)	−.015 [.016]
Share of HHs with electric lighting	.46 (.32)	.67 (.28)	−.018 [.016]
Share of HHs with electric fan	.28 (.22)	.42 (.29)	−.019 [.017]
Share of HHs with TV	.20 (.14)	.30 (.20)	−.025** [.012]
Share of HHs with refrigerator	.02 (.05)	.07 (.11)	−.004 [.004]
Share of HHs with air conditioning	.03 (.05)	.04 (.09)	−.006 [.005]
Number of districts	229	332	

NOTE.—Panel A reports means and standard deviations of village-level covariates from the 2001 census, the 2005 Economic Census, and 2005–6 DISE school data. All three columns include only villages with 2001 populations between 150 and 450, which is slightly wider than our optimal RD bandwidths. Column 2 includes districts in the first wave of RGGVY implementation. Column 3 further restricts the sample to single-habitation villages in 10th-Plan districts, in states with reliable village shapefiles. Panel B reports district-level means and standard deviations for 10th- versus 11th-Plan districts using the 2005 NSS (representative at the household [HH] level). Column 3 reports district-level pretrend estimates using 2000 and 2005 NSS data, comparing the 10th and 11th Plans, 2005 vs. 2000 (including state-specific linear trends; standard errors are in brackets). Tables B7 and B8 report these regression results in full.

** $p < .05$.

light intensity, on a 0–63 scale, at approximately 1-km² resolution. We construct a yearly brightness panel by assigning each village the maximum brightness over all pixels in its shapefile polygon.²⁴ We are unable to construct village-level brightness in 10 states for which 2001 village shapefiles are unavailable or unreliable.²⁵ We exclude these states from our RD analysis in order to control for pre-RGGVY brightness at the village level, which is important for statistical power; our DD analysis includes these 10 states, since fixed effects subsume baseline controls (see eq. [2]).

Satellite-derived brightness proxies for electrification in small villages, as it is proportional to observed luminosity from electric lighting (Chen and Nordhaus 2011).²⁶ These data capture lighting consumption from all users (including households and enterprises), providing a lower bound for total power consumption—electric groundwater pumps share the same power grid but do not emit light. They also capture power from all sources, meaning that any estimated brightness effects are net of substitution between off-grid and grid power. One caveat is that these data cannot distinguish between changes in street lighting and broader gains in electricity access. However, RGGVY did not install streetlights in the 10th Plan. Therefore, any RGGVY-driven increases in nighttime brightness are likely to reflect village-wide expansions in access to energy services.

Finally, we construct a repeated cross section from the “thick” 2000, 2005, and 2010 waves of India’s National Sample Survey (NSS). Each wave surveyed 60,000–80,000 rural households, with sampling weights that are representative at the district level. While NSS data lack the village identifiers required for our RD analysis, they directly report household-level electricity consumption, appliance ownership, and total expenditure.²⁷ We collapse these data into a three-wave district-level panel, which we use in our DD analysis. We use two NSS measures of rural electricity access: (i) the share of households consuming any electricity and (ii) monthly kilowatt-hours

²⁴ Indian villages are organized into central clusters of households surrounded by agricultural fields. Assigning villages their maximum brightness targets our electricity proxy on populated areas, rather than unlit cropland. We remove year-specific measurement error in brightness via linear projection, while also dropping a few extreme outliers. Appendix C.3 provides further detail on our nighttime brightness data.

²⁵ Shapefiles are unavailable for Arunachal Pradesh, Meghalaya, Mizoram, Nagaland, and Sikkim. For Assam, Himachal Pradesh, Jammu and Kashmir, Uttar Pradesh, and Uttarakhand, available shapefile polygon areas are uncorrelated with village areas reported in the 2001 census. Our remaining sample states contain 60% of 10th-Plan RGGVY villages. See app. C.2 for further discussion.

²⁶ Several remote-sensing studies have ground-truthed the relationship between nighttime brightness and electrification in villages in India (Min 2011), South Africa (Machemedze et al. 2017), Vietnam (Min and Gaba 2014), and Senegal and Mali (Min et al. 2013).

²⁷ NSS electricity consumption data capture all power purchases, including both grid and off-grid sources (e.g., kilowatt-hours purchased from local diesel generator operators). A separate NSS variable reports diesel and petrol purchases for the purpose of self-generation (distinct from fuels purchased for transportation). We find no statistical evidence that RGGVY crowded out self-generated electricity (see table A11).

TABLE 2
VILLAGE-LEVEL RD IN 2011 ELECTRICITY ACCESS, BY SECTOR

	OUTCOME: VILLAGE-LEVEL ELECTRICITY ACCESS			
	Domestic (1)	Agricultural (2)	Commercial (3)	All 3 Sectors (4)
A. Dummy for Any Power Access				
1[2001 population ≥ 300]	−.004 (.010)	−.001 (.020)	.043*** (.017)	.038** (.016)
Mean of dependent variable (<300)	.906	.669	.436	.417
Optimal bandwidth	108	78	136	150
Village observations	13,517	9,836	16,900	18,574
B. Hours/Day of Power Supply				
1[2001 population ≥ 300]	−.042 (.207)	−.252 (.257)	.555** (.220)	.283 (.240)
Mean of dependent variable (<300)	11.386	5.382	3.957	5.050
Optimal bandwidth	88	82	126	117
Village observations	9,284	8,575	12,897	14,336

NOTE.—All estimates, computed using rdrobust, use linear polynomials, triangular kernels, MSE-optimal bandwidth, and nearest-neighbor standard errors. Regression samples include within-bandwidth single-habitation villages in RGGVY 10th-Plan districts (i.e., the first wave of RGGVY project implementation, for which 300 people is the relevant 2001 population-based eligibility cutoff). Each regression controls for pre-2005 nighttime brightness at the village level and state fixed effects. Optimal bandwidths are symmetric above and below the 300-person cutoff, and we report means of the dependent variable for villages below the cutoff. In panel A, outcomes are dummy variables for electricity access at the village level. In panel B, outcomes are the average hours of power available per day in the village. Results are robust to alternative controls, kernels, bandwidth algorithms, and standard errors.

** $p < .05$.
*** $p < .01$.

consumed by the representative household. Our primary NSS outcome variable is monthly expenditure per capita, which aligns with SHRUG’s village-level expenditure variable. This follows a tradition of using consumption spending to approximate well-being in development economics.²⁸ Panel B of table 1 reports NSS summary statistics before RGGVY, comparing 10th- and 11th-Plan districts.

V. Impacts of RGGVY

A. First Stage: Electricity Access and Consumption

Village-level electricity access.—Using our RD strategy, we estimate RGGVY’s impact on two sets of electricity outcomes from the 2011 census: (i) dummies for a village’s first grid connection to an end-use sector and (ii) average hours per day of power supply by sector. Table 2 reports RD estimates

²⁸ Examples include Banerjee et al. (2015), Haushofer and Shapiro (2016), Topalova (2010), and Atkin et al. (2024). The latter two papers both use this same NSS expenditure variable. Appendix C.11 provides more details.

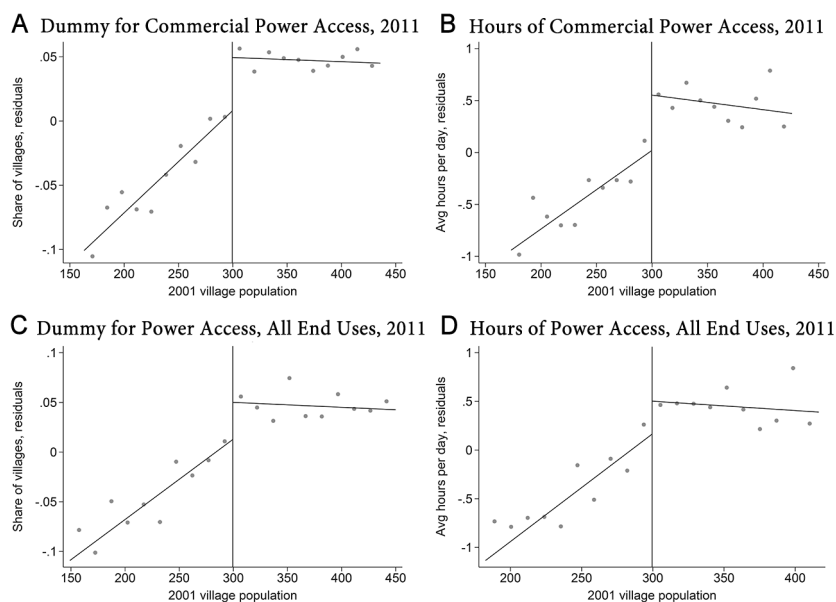


FIG. 4.—Village-level RDs in 2011 electricity access. *A* and *B* correspond to column 3 of table 2. *C* and *D* correspond to column 4 of table 2. See table notes for details. Figure A4 displays RD plots corresponding to the other four regressions in table 2.

for 300-person villages, and figure 4 presents the corresponding RD plots.²⁹ We find that RGGVY eligibility caused a 3.8-pp increase in the share of villages reporting all-sector electricity access—a 9% increase over the baseline mean, statistically significant at the 5% level. This is driven by a 4.3-pp (10%) increase in commercial power access, statistically significant at the 1% level. We find no impacts on the extensive margin of domestic (−0.4 pp) or agricultural (−0.1 pp) power access. This is consistent with RGGVY’s emphasis on more intensively electrifying villages where the first household already had a grid connection. By contrast, the program’s goal to support microenterprises appears to have connected 1 in 13 barely eligible villages that previously lacked any commercial power access.³⁰

We find that RGGVY eligibility increased commercial power supply by 0.56 hours per day (14%), statistically significant at the 1% level. This intensive-margin increase implies that newly connected villages received

²⁹ Figure A4 shows RD plots for the domestic and agricultural sectors, omitted here for brevity. Table A1 presents “difference-in-discontinuities” results, which are quantitatively similar.

³⁰ With 44% of barely ineligible villages having commercial power access, the maximum increase we could estimate would be 56 pp. Our null effects for domestic access are unsurprising, given that 91% of barely ineligible villages had domestic power.

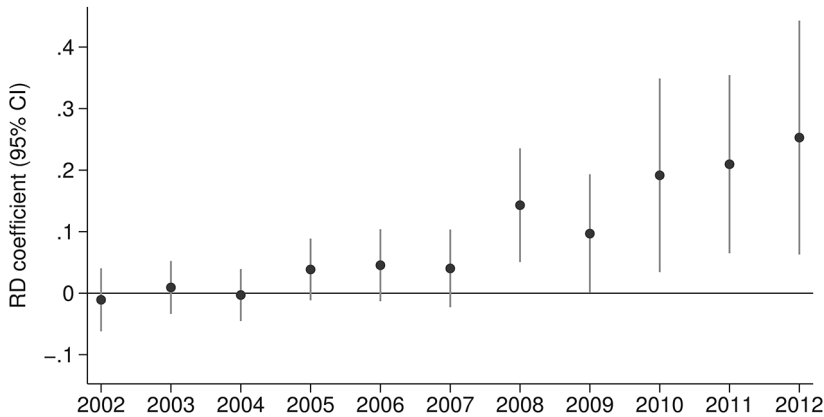


FIG. 5.—Village-level RD estimates in nighttime brightness, by year. This figure plots RD coefficients for maximum nighttime brightness at the village level. We estimate a separate regression for each year, with *rdrobust* specifications identical to those in table 2. Each regression controls for pre-2005 brightness at the village level; 2002–5 regressions control for brightness in years preceding the outcome variable. Optimal bandwidths for these regressions range from 69 to 143. Results are robust to alternative controls, kernels, bandwidth algorithms, and standard errors. See the table 2 note for details. Tables A10 and B5 report these results numerically. Whiskers display 95% confidence intervals (CIs).

13 hours per day of commercial power supply, above the median of 10 hours per day for electrified villages in our RD sample. We do not detect a similar increase in hours of all-sector power supply, which likely reflects noncoincident consumption patterns across electricity end uses (Burgess et al. 2023). Table A2 uses our DD strategy to estimate impacts on village-level power access beyond the RD bandwidth. This reveals a near-identical effect (3.7 pp) on all-sector power access, with larger increases for the domestic (8.1 pp) and agricultural (5.2 pp) sectors. As further evidence of RGGVY's impact on households, we estimate a 3.7-pp increase in the share of households using electricity as a main source of lighting, applying our RD strategy to 10th-Plan districts with high RGGVY treatment intensity (see table A13).³¹

Nighttime brightness.—Figure 5 presents RD estimates of the effect of RGGVY eligibility on nighttime brightness. We find null results before RGGVY's 2005 announcement, consistent with our RD assumption of baseline covariate smoothness.³² Our RD estimates increase almost monotonically each year thereafter and are statistically different from zero by 2008. By 2012, 4–6 years after 10th-Plan districts received funding, RGGVY eligibility had increased brightness by 0.25 units at the 300-person

³¹ We discuss this analysis of heterogeneous treatment intensity in sec. VI.B below.

³² Figures A5 and B6 present the corresponding RD plots. Tables A10 and B5 report these results numerically. We find similar results if we estimate a single “difference-in-discontinuities” regression, rather than separate regressions for each year (see fig. A1).

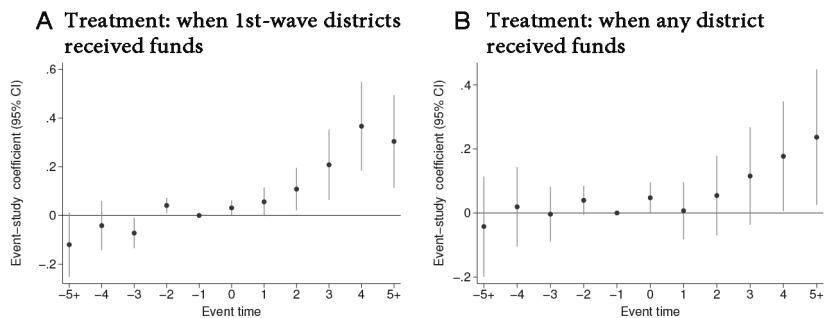


FIG. 6.—Village-level DD event studies using annual nighttime brightness from 1998 to 2013. The outcome variable is maximum brightness in each year, for each village polygon. In *A*, treatment (RGGVY eligibility) turns on in the year when each 10th-Plan district first received RGGVY project funds, using 11th-Plan districts as controls. In *B*, treatment turns on for both 10th- and 11th-Plan districts, in the first year the district received RGGVY funds. The 10th-Plan districts first received funds in 2005–7, while 11th-Plan districts first received funds in 2008–11. In both panels, the omitted year is the last year before a district’s first receipt of funds. Regressions include village fixed effects, state-by-year fixed effects, and village-specific linear time trends. Estimation samples include 10th- and 11th-Plan districts in states with reliable shapefiles, without restricting village size. Whiskers display 95% confidence intervals (CIs), with standard errors clustered by census block.

cutoff (statistically significant at the 1% level). Yearly brightness data also allow us to estimate DD event studies, where treatment turns on in the year district *d* first received RGGVY funds. Figure 6 reports these results, which align with our RD findings: 4–5 years after RGGVY project funding, average village brightness had differentially increased by 0.18–0.37 units.³³ As with our RD results, our event studies find no evidence of differential brightness before RGGVY treatment.

To interpret these effect sizes, we use estimates from the remote-sensing literature that ground-truth the relationship between rural electrification and nighttime brightness. Min et al. (2013) find that electrification is associated with a 0.36-unit increase in brightness in rural villages in Senegal. Min and Gaba (2014) find that a 1-unit increase in brightness corresponds to 60 public streetlights or 240–270 electrified homes in Vietnamese villages. Finally, Machemedze et al. (2017) find that connecting 50 South African homes to the grid is associated with a 0.35-unit brightness increase. In the context of these estimates, our results suggest that RGGVY caused meaningful on-the-ground increases in electricity use: a 0.25-unit brightness increase is associated with a roughly 10-pp increase in the share of households with electric lighting, net of any substitution away from off-grid power sources.³⁴

³³ This leverages both RGGVY’s staggered rollout across the 10th and 11th Plans and the staggered timing of DPR funding within each plan, letting us capture RGGVY’s delayed impacts in 11th-Plan “control” districts (fig. 6*B*). Table A4 reports analogous pooled DD estimates.

³⁴ See fig. A14, which also shows that a village’s nighttime brightness is positively correlated with its hours per day of commercial power supply. Our nighttime brightness effects

Household electrification.—Using our DD design with NSS data, we can directly estimate RGGVY's impacts on households. Table 3 presents these results. We find a 5.6-pp increase on the extensive margin of household electricity consumption (statistically significant at the 1% level); this intent-to-treat effect implies a 9% increase in household grid connections. We also find increases on the intensive margin of household electricity consumption, with intent-to-treat effects of 4 kWh per month (13%, statistically significant at the 5% level). We find corresponding increases of 4.9 pp (8%) in electric lighting and 4.5 pp (11%) in electric fan adoption but null effects for more expensive appliances.³⁵

These NSS estimates corroborate our village-level RD results and demonstrate that RGGVY had a meaningful impact on household electricity consumption. Our 5.6-pp extensive-margin estimate implies that RGGVY connected 14% of previously unelectrified households in 10th-Plan districts.³⁶ Our 4-kWh intensive-margin estimate is consistent with moving all newly connected households from 0 kWh to (above) the mean consumption level of electrified households. Our results also confirm that RGGVY fell short of full electrification. Since our first-stage estimates are robust and statistically precise, we can scale up RGGVY program impacts to estimate the effects of electrification on development outcomes.

First-stage robustness.—Figures B1 and B2 show that our first-stage RD results are broadly robust to alternative weighting kernels, bandwidth algorithms, standard errors, fixed effects, polynomials, and sample criteria. While a few sensitivities reduce statistical precision (for all-sector power access, in particular), our RD point estimates are stable across these variants. We also conduct RD falsification tests using samples of (i) 11th-Plan villages, for which the 300-person cutoff did not determine RGGVY eligibility, and (ii) multihabitation villages, for which total village population is the wrong running variable (since RGGVY's eligibility threshold was based on habitation population). Figures B3 and B4 reveal null effects for all but the correct RD sample of 10th-Plan single-habitation villages.³⁷

should reflect changes in power use across sectors of the village economy (including households and shops that operate at night).

³⁵ Table A5 repeats this DD analysis using the analogous census variables, finding a similar 3.4-pp lighting effect. Our main DD specifications include state-specific trends, which are fitted through our three NSS waves only. We find similar results when we remove these trends (see table A6).

³⁶ The 95% confidence interval for col. 2 of table 3 includes an 8.2-pp increase, which is consistent with RGGVY having electrified one in five previously unconnected households (Sreekumar and Dixit 2011).

³⁷ Our RD results pass placebo tests that compare the correct 300-person cutoff to a distribution of randomly generated cutoffs (see fig. B5). Our NSS results pass an analogous randomization test that scrambles the assignment of districts into the 10th-Plan "treated" group (see fig. B9).

TABLE 3
DISTRICT-LEVEL DD OF HOUSEHOLD ELECTRICITY ACCESS AND USAGE

	HOUSEHOLD ELECTRICITY USE (kWh/month)			1[HOUSEHOLD OWNS ELECTRIC APPLIANCE]				
	1[Q > 0] (1)	Levels (2)	Logs (3)	Lighting (4)	Fan (5)	TV (6)	Refrig. (7)	AC (8)
1[10th-Plan] × 1[2010]	.056*** (.014)	3.95** (1.75)	.171** (.075)	.049*** (.015)	.045*** (.016)	.010 (.014)	.002 (.007)	.007 (.005)
Mean of dependent variable	.590	31.45	3.038	.598	.382	.289	.055	.038
Clusters	552	552	550	552	552	552	552	552

NOTE.—District-level DD with three NSS years (2000, 2005, 2010). We aggregate household-level data up to the district, using sampling weights, for rural households only. Outcome variables are an indicator for whether a household consumed any electricity (col. 1), monthly household electricity consumption in levels and in natural logs (cols. 2–3), and indicators for whether a household owned electric lighting, an electric fan, a television, a refrigerator, or air conditioning (AC; cols. 4–8). DD treatment is assigned at the district level, for 10th-Plan districts. All regressions include district fixed effects; year fixed effects; state-specific linear trends; linear trends in state quartiles of 2005 household expenditures per capita, to control for within-state selection in RGGVY implementation based on relative differences between districts (e.g., states prioritizing electrification in their poorest districts); and linear trends in national deciles of 2005 household expenditures per capita control for such selection in absolute terms. Standard errors are clustered at the district level, collapsing to a single cluster for (i) districts that split in the 2001 census but that the 2000 NSS sampled on the basis of 1991 district definitions and (ii) the few cases where an RGGVY DPR included multiple districts. For all columns, the number of observations is 1,670.

** $p < .05$.
*** $p < .01$.

B. *Reduced Form: Economic Outcomes*

Having established that RGGVY increased rural electrification, we perform a “program evaluation” by estimating local intent-to-treat effects on a range of development outcomes. Table 4 reports these village-level RD estimates, while figure 7 presents the corresponding RD plots for key outcomes.³⁸ We estimate precise null effects for our preferred economic outcome, SHRUG expenditure per capita (table 4, panel A). We can reject increases greater than Rs 29 per month (2% of mean expenditure). We can also reject over-3-pp decreases in the share of households with a poverty indicator and over-1-pp increases in the share of low-income households with a salaried job. In panel B, we test for the Tiebout (1956) “vote-with-their-feet” mechanism, by using 2011 population as an

³⁸ Figures A10–A12 present RD plots for other outcomes in table 4, omitted here for brevity. We report reduced-form RD sensitivities in tables B1–B4; our results are similar under alternative kernels and RD bandwidths. Table A22 reports RD results for additional village-level outcomes, including housing characteristics and community-wide amenities. Panel L presents indexed and aggregated outcomes, to address any multiple-testing concerns.

TABLE 4
VILLAGE-LEVEL RD: REDUCED-FORM OUTCOMES

	RD Estimate	Standard Error	95% Confidence Interval	Mean Y_v
A. Consumption and Income (2011)				
Expenditure per capita (Rs/month)	-5.222	(17.565)	[-39.649, 29.206]	1,365.353
Expenditure per capita (logged)	-.010	(.013)	[-.034, .015]	9.668
Share HHs with poverty indicator	-.004	(.013)	[-.030, .022]	.547
Share HHs that rely on cultivation	-.007	(.012)	[-.030, .016]	.421
Share HHs earning >Rs 5,000/mo	.002	(.009)	[-.016, .020]	.070
Share HHs with salaried job	.004	(.003)	[-.003, .010]	.012
B. Village Demographics (2011)				
Population	6.213	(3.874)	[-1.379, 13.805]	296.447
Share population age 0-6	.001	(.002)	[-.002, .004]	.141
Average HH size	.024	(.024)	[-.023, .072]	4.908
C. Workers as Share of Population (2011)				
Agricultural workers:				
Total	-.006	(.007)	[-.019, .007]	.399
Male	-.007	(.006)	[-.018, .004]	.465
Female	-.005	(.009)	[-.024, .013]	.329
Nonagricultural workers:				
Total	.004	(.003)	[-.002, .011]	.075
Male	.004	(.004)	[-.005, .013]	.096
Female	.005	(.004)	[-.003, .013]	.053
D. Firm Outcomes (2013)				
No. of firms	.812	(.716)	[-.591, 2.214]	8.125
No. of firm employees	-2.173	(4.620)	[-11.228, 6.882]	15.969
E. School Outcomes (2014-15 School Year)				
No. of students enrolled:				
Grades 1-5	3.086	(3.681)	[-4.128, 10.301]	46.417
Grades 6-8	-1.949	(2.394)	[-6.642, 2.744]	10.314
No. of students enrolled:				
Grades 4-5	-.438	(.510)	[-1.437, .561]	5.150
Grades 7-8	-.558	(.418)	[-1.378, .261]	1.469

NOTE.—Each row reports results from a separate RD regression. HH = household. In panels B and C, we control for the 2001 level of the outcome variable. In panels D and E, we control for the 2005 (or 2005-6) level of the outcome variable. RD-robust regressions are otherwise identical to those in table 2. Optimal bandwidths range from 71 to 136 above/below 300 people. Results are broadly robust to alternative controls, kernels, bandwidth algorithms, and standard errors. The rightmost column reports means of the outcome variable for villages below the 300-person cutoff.

RD outcome. While we find a positive point estimate of 6.2 people, we can reject increases greater than 14 people (5%).

Panels B-E of table 4 present additional reduced-form RD results for village demographics, employment, firms, and education. We find no statistical evidence that RGGVY eligibility caused nonzero impacts at the 300-person cutoff in any of these outcomes. Null results for firm and

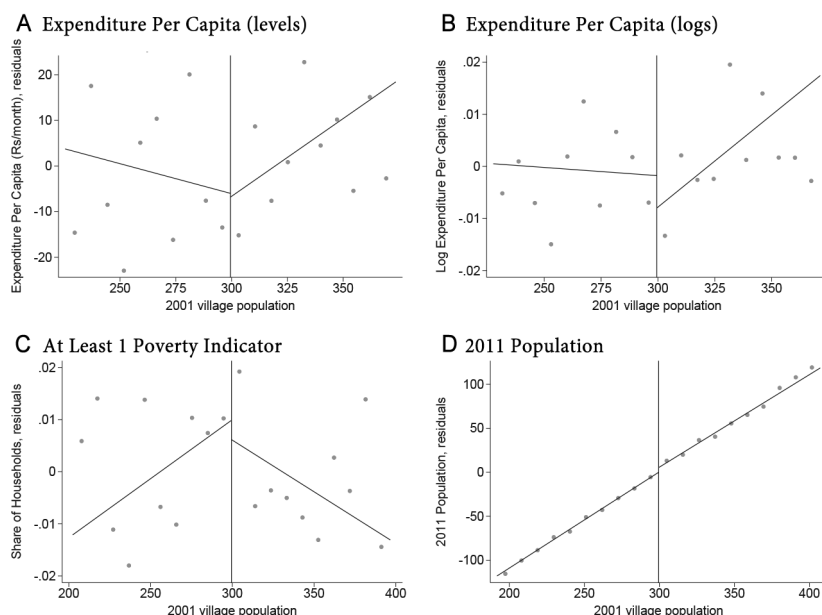


FIG. 7.—Village-level RDs in expenditure, poverty, and population. RD plots correspond to four regressions in table 4: rows 1–3 of panel A, and row 1 of panel B. See table notes for details. Figures A10–A12 display RD plots for the other regressions in table 4.

education outcomes have the least precision, but these point estimates still indicate relatively small effects.³⁹

Table 5 presents reduced-form DD results for NSS per capita expenditure.⁴⁰ As with our SHRUG expenditure RD, we find precise null results. For our point estimate of Rs 27 per month (3% of mean expenditure), we can reject RGGVY-induced increases greater than Rs 75 per month (8%) for the average household.

Taken together, our RD and DD results imply that while the RGGVY program meaningfully increased electricity access and consumption, it yielded far smaller short- to medium-run economic improvements than promised by policy makers.⁴¹ To properly interpret these magnitudes,

³⁹ We report the analogous DD estimates in table A23. They are broadly consistent with our RD results in table 4, except for statistically significant increases in household size (but not fertility) and school enrollment (not robust to a DD analysis that includes RGGVY's 11th-Plan rollout).

⁴⁰ Before aggregating total expenditures to the district level (using NSS sampling weights), we subtract spending on electricity. Net expenditures are a more appropriate welfare proxy, since they account for the benefits and the costs of consuming electricity.

⁴¹ For example, Ministry of Power (2013, 2) explains that “RGGVY was launched with the objective that it would transform the quality of life in rural India and make possible the

TABLE 5
DISTRICT-LEVEL DD OF HOUSEHOLD CONSUMPTION EXPENDITURES

	EXPENDITURE PER CAPITA (Rs/month)	
	Levels (1)	Logs (2)
$I[10\text{th-Plan}] \times I[2010]$	27.47 (24.05)	.029 (.022)
Mean of dependent variable	978.15	6.833
Clusters	552	552

NOTE.—District-level DD with three NSS years (2000, 2005, 2010). The outcome variable is total household expenditures per capita (net of electricity spending per capita) over the 30-day period before survey enumeration, in 2010 rupees per month (col. 1) and log transformed (col. 2). Both regressions include district fixed effects, year fixed effects, state-specific linear trends, linear trends in state quartiles of 2005 household expenditures per capita, and linear trends in national deciles of 2005 household expenditures per capita. See table 3 note for further details. Standard errors are clustered at the district level, collapsing to a single cluster for (i) districts that split in the 2001 census but that the 2000 NSS sampled on the basis of 1991 district definitions and (ii) the few cases where an RGGVY DPR included multiple districts. For all columns, the number of observations is 1,670.

section VI rescales our estimates to quantify the benefits of “full electrification” and conducts a welfare analysis by comparing these benefits to the costs of electrification.

VI. Economic Impacts of Electrification

Next, we move beyond the RGGVY program to estimate the development impacts of rural electrification more broadly. Since RGGVY did not electrify all villages and rural households, it is possible that a more expansive “full-electrification” program would have yielded more pronounced economic impacts. To bridge this gap, we scale our reduced-form estimates by our first-stage estimates of RGGVY’s impacts on electricity access and consumption.

A. Rescaled Treatment Effects

We implement a village-level fuzzy RD using two endogenous treatment variables: daily hours of commercial power supply and nighttime brightness. For each treatment variable, we apply an additional scaling factor in order to interpret the resulting estimates relative to a “full-electrification” benchmark. We scale commercial power supply by 10 hours per day—the median 2011 supply in villages with nonzero commercial access.⁴² This

attainment of 8% economic growth and bridging the urban-rural gap in terms of quality of life.”

⁴² A 10-hour increase corresponds to providing new commercial connections at the median power quality or to shifting a previously connected village from the 25th to the

places a realistic limit on “full electrification” in a context where less than 5% of electrified villages received 24-hour power supply. We scale nighttime brightness by 2.6 units, corresponding to a shift from the 25th to the 75th percentile of 2011 brightness in our RD sample.⁴³

Table 6 presents fuzzy RD results for our preferred village-level outcome, expenditure per capita. Panel A uses hours of commercial power as an endogenous variable, and we find that 10 additional hours caused a statistically insignificant decrease of Rs 222 per month. Panel B uses nighttime brightness as the “treatment” variable, and we find that a 2.6-unit brightness increase caused a statistically insignificant decrease of Rs 96 per month. Both rescaled estimates yield similar upper confidence intervals: we can reject LATEs greater than Rs 347 and Rs 359 per month (25% and 26%, respectively). These negative point estimates imply that “full electrification” is unlikely to create large expenditure increases in 300-person villages in the short-to-medium term.⁴⁴

Next, we scale up our reduced-form DD estimates using two-stage least squares, instrumenting with whether each district belonged to RGGVY’s 10th Plan. Here, our endogenous NSS “treatment” variable is the share of households consuming any electricity. We report these results in columns 1 and 4 of table 7. Our statistically insignificant point estimate suggests a per capita expenditure increase of Rs 316 per month (32%); we cannot reject that a 100-pp increase in household connections would double expenditures.⁴⁵

These DD-IV estimates differ from our fuzzy RD LATEs, likely because they average over villages of all sizes: the 50th- (90th-)percentile NSS village had a population of 1,913 (6,854). To make our DD-IV and fuzzy RD estimates more comparable, while also reflecting the reality of last-mile electrification, we need to exclude very large villages from the NSS sample when scaling up to “full electrification.”⁴⁶ We construct separate district-level panels for quintile 1 (Q1) and quintiles 2–5 (Q25) of village sizes,

75th percentile of commercial power quality. Figure A13 presents this distribution (mean of 10.9 hours, interquartile range of 9 hours).

⁴³ Figure A13 presents this distribution of 2011 nighttime brightness for our RD sample (mean of 6.2, interquartile range of 2.6). Figure A14 shows that nighttime brightness is positively correlated with both the household penetration of electric lighting and hours of commercial power at the village level.

⁴⁴ Tables A30 and A31 report the analogous fuzzy RD results for the remaining outcomes in table 4, all of which are statistically indistinguishable from zero.

⁴⁵ While this F -statistic of 11.9 meets the $F \geq 10$ “rule of thumb,” it is less than the Stock and Yogo (2005) critical value of 16.74. Table B11 presents the analogous ordinary least squares regressions (without instrumenting).

⁴⁶ Compared to sub-500-person villages, villages with over 4,000 people were twice as likely to have all-sector power access before RGGVY. Given greater preexisting power access, large villages are also less externally valid for considering a 0–100-pp increase in household connections (i.e., “full electrification”).

TABLE 6
FUZZY RD IN EXPENDITURE PER CAPITA, USING TWO ENDOGENOUS VARIABLES

	EXPENDITURE PER CAPITA (Rs/month)	
	Levels (1)	Logs (2)
A. Hours/Day of Commercial Power		
Fuzzy RD point estimate	−22.160 (29.019)	−.020 (.021)
LATE for a 10-hour increase	−221.603	−.203
95% confidence interval for a 10-hour increase	[−790.4, 347.2]	[−.613, .208]
Mean of dependent variable (<300)	1,366.7	9.668
Optimal bandwidth	106	115
Village observations	10,402	11,240
B. Units of Nighttime Brightness		
Fuzzy RD point estimate	−36.900 (89.315)	−.054 (.066)
LATE for a 2.6-unit increase	−95.940	−.140
95% confidence interval for a 2.6-unit increase	[−551.1, 359.2]	[−.476, .496]
Mean of dependent variable (<300)	1,364.3	9.667
Optimal bandwidth	99	102
Village observations	11,716	12,045

NOTE.—Fuzzy RD robust estimates using two endogenous village-level “treatment” variables: 2011 average hours per day of commercial power in cols. 1–2 and 2011 nighttime brightness in cols. 3–4. For hours of commercial power, we scale up by a factor of 10, the median of the 2011 distribution of nonzero hours of commercial power in the RD bandwidth. For nighttime brightness, we use a scaling factor of 2.6, equal to the interquartile range of the 2011 distribution of village-level brightness in the RD bandwidth. Both scaling factors denominate LATEs for a village moving from the 25th to the 75th percentile of the “treatment” variable. Outcomes variables are 2011 SHRUG consumption expenditures per capita in levels (Rs/month) and logs. Each fuzzy RD uses a linear polynomial, a triangular kernel, an MSE-optimal bandwidth, and nearest-neighbor standard errors. Regression samples include within-bandwidth single-habitation villages in RGGVY 10th-Plan districts (i.e., the first wave of RGGVY implementation, for which 300 people is the relevant eligibility cutoff). Regressions control for pre-2005 nighttime brightness at the village level and state fixed effects. Optimal bandwidths are symmetric above and below the 300-person cutoff, and we report means of the dependent variable for villages below the cutoff. Results are robust to alternative controls, kernels, and bandwidth algorithms.

proxying for population using village-level NSS sampling weights. Isolating Q1 villages (with small sampling weights) removes the vast majority of villages over 4,000 people, shifting the NSS distribution toward smaller villages with a median population of 1,043.⁴⁷ Q1 villages show large first-stage

⁴⁷ Ideally, we would split the sample using village populations; however, we observe populations for only the 2005 and 2010 NSS waves. We prefer to split using sampling weights, retaining the 2000 NSS wave for statistical power. This aligns with the NSS’s sampling strategy, which is explicitly proportional to village size. Figure C6 plots village populations in the 2005 and 2010 NSS waves, split by Q1 vs. Q25, showing that splitting on sampling weights is an effective way of removing large villages from the sample.

TABLE 7
DISTRICT-LEVEL DD-IV OF HOUSEHOLD CONSUMPTION EXPENDITURES

	EXPENDITURE PER CAPITA (Rs/month)					
	Levels			Logs		
	(1)	(2)	(3)	(4)	(5)	(6)
1[HH consumes any electricity]	315.9 (529.8)	-680.4 (518.7)	887.2 (646.7)	.361 (.422)	-.229 (.327)	.629 (.518)
95% confidence interval	[-724.8, 1,356.5]	[-1,704.7, 343.8]	[-383.4, 2,157.9]	[-.469, 1.190]	[-.875, .417]	[-.389, 1.646]
Village weight quintiles	Pooled	1	2-5	Pooled	1	2-5
50th percentile of 2001 population	1,913	1,043	2,076	1,913	1,043	2,076
90th percentile of 2001 population	6,854	4,875	7,291	6,854	4,875	7,291
Mean of dependent variable	978.2	1,128.2	948.2	6,833	6,957	6,804
Clusters	552	162	494	552	162	494
Observations	1,670	418	1,488	1,670	418	1,488
First-stage estimate	.048***	.202***	.043***	.048***	.202***	.043***
(standard error)	(.014)	(.049)	(.015)	(.014)	(.049)	(.015)
First-stage <i>F</i> -statistic	11.91	16.75	8.84	11.91	16.75	8.84

NOTE.—District-level DD with three NSS years (2000, 2005, 2010), estimated via two-stage least squares. We instrument for household electricity access with the interaction 1[10th-Plan district] $\times$ 1[2010]. The outcome variable is net of per capita spending on electricity. Columns 2-3 and 5-6 split the sample on within-year quintiles of NSS village weights before collapsing to the district level using sampling weights. We do not observe village populations for the 2000 NSS wave, meaning that we cannot estimate a three-period panel splitting directly on village size. However, isolating the first quintile of NSS sampling weights shifts the distribution of 2001 village populations (as observed in 2005 and 2010 waves) toward smaller villages. All regressions include district fixed effects, year fixed effects, linear trends in state quartiles of 2005 household expenditures per capita, and linear trends in national deciles of 2005 household expenditures per capita. See table 3 note for further details. Standard errors are clustered at the district level, collapsing to a single cluster for (i) districts that split in the 2001 census but that the 2000 NSS sampled on the basis of 1991 district definitions and (ii) the few cases where an RGCVY DPR included multiple districts. The bottom three rows report the first-stage point estimates and standard errors and Kleibergen-Paap first-stage *F*-statistics.

*** $p < .01$.

effects, consistent with smaller villages having had fewer household connections before RGGVY.⁴⁸

Columns 2 and 5 of table 7 present DD-IV results for the Q1 subsample. Our point estimate indicates a decrease in per capita expenditure of Rs 680 per month (not statistically distinguishable from zero), and we can reject increases from “full electrification” greater than Rs 344 per month (30%).⁴⁹ The upper confidence intervals from our Q1 DD-IV estimates and our fuzzy RD LATEs are very similar—despite using different consumption measures, identification strategies, and endogenous treatment variables.

Our results for larger villages tell a different story (cols. 3 and 6 of table 7). Our point estimates for the Q25 subsample are positive and large but statistically imprecise. We cannot reject a tripling of expenditures in these larger villages. When we further restrict the Q25 subsample to districts with high power quality, we estimate a per capita expenditure increase of Rs 1,428 per month (139%, statistically significant at the 10% level).⁵⁰

How should we interpret these magnitudes? We are estimating treatment effects of a 100-pp increase in electricity access; shifting from a 0% to a 100% grid connection rate has the potential to dramatically alter production and consumption in the village economy. Nevertheless, in small villages, we find that last-mile electrification does not lead to expenditure gains on average. In contrast, we estimate nearly a doubling of per capita expenditure in large villages, consistent with a prior literature that has found large impacts of electrification.⁵¹ To fully evaluate the welfare impacts of rural electrification, we compare these benefits to their costs in section VI.C.

B. Heterogeneous RGGVY Implementation Intensity

A potential concern with scaling up to “full-electrification” treatment effects is that doing so extrapolates beyond the support of the RGGVY

⁴⁸ Tables A8 and A9 report first-stage estimates for both Q1 and Q25 subsamples. Table A7 finds similar extensive-margin results for a two-wave NSS panel, comparing villages larger and smaller than 2,000 people (roughly the median NSS village size). The sub-2,000-person subsample produces results similar to the Q1 subsample, but with a weaker first-stage *F*-statistic. Figure A2 replicates these DD population splits using a 2-year census panel, for the extensive margin of village (rather than household) electricity access. This likewise reveals larger first-stage effects for smaller villages.

⁴⁹ Table A29 reports DD-IV results for a two-wave NSS panel of sub-2,000-person villages; these estimates are very similar to our Q1 estimates in table 7 (though splitting directly on village population shortens our NSS panel to two waves, weakening the first stage). Our DD-IV regressions also lose first-stage power if we include state-specific trends (see table B10), yet the Q1 point estimates remain similar.

⁵⁰ Table A21 reports this result (with a first-stage *F*-statistic of 15.8). By contrast, when we impose this same restriction on the Q1 subsample, we can reject 26% increases in expenditure per capita.

⁵¹ For example, IV point estimates in Dinkelman (2011) imply that a 100-pp increase in electrification would double male earnings.

policy. To address this concern, we leverage heterogeneity in RGGVY's implementation intensity across 10th-Plan districts. We isolate the 90 "high-intensity" districts where RGGVY treated at least 60% of villages, thereby coming closest to the "full-electrification" ideal.⁵² In these districts, our first-stage RD coefficients nearly double, to 0.973 hours of commercial power access and 0.41 units of nighttime brightness. Despite these much stronger first-stage effects, our reduced-form RD results are largely unchanged.⁵³ However, our NSS DD estimates for the Q25 subsample reveal 9% reduced-form expenditure increases in high-intensity districts (statistically significant at the 1% level). Figure 8 presents both sets of reduced-form expenditure results, demonstrating the stark contrast between the null expenditure effects in 300-person villages and large expenditure gains in larger villages.

C. *Welfare Analysis*

Finally, we evaluate the welfare effects of "full electrification." We use two strategies for quantifying economic benefits, which we compare against the costs of electrification. First, we use our fuzzy RD and DD-IV estimates of the effects of electrification on expenditure per capita. This welfare proxy is well suited to measure treatment effects from an electrification program that affects multiple sectors of the village economy, since consumption spending should capture any electricity-induced changes in agricultural and firm productivity.⁵⁴ To account for the uncertainty in our econometric estimates, we take 10,000 draws from the sampling distributions of our fuzzy RD coefficients in column 1 of table 6 and our DD-IV coefficients in columns 2 and 3 of table 7.⁵⁵

We compute village surplus by multiplying per capita expenditure by population size, to measure effects for 300- (aligning with our fuzzy RD LATEs), 1,000-, and 2,000-person (roughly the median village sizes in

⁵² Figure A6 shows the distribution of RGGVY implementation intensity across 10th-Plan districts. In 35 "low-intensity" 10th-Plan districts, RGGVY treated less than 60% of eligible villages. We report the results of this heterogeneity analysis in tables A12–A17 and fig. A7.

⁵³ While we find suggestive evidence that RGGVY may have decreased the share of households with poverty indicators by 3.3 pp or increased nonagricultural employment by 0.9 pp (both statistically significant at the 10% level), these magnitudes remain modest in high-intensity districts (see table A15).

⁵⁴ The median village in the 2013 Economic Census had 1.7 employees per firm. Any electricity-induced productivity gains for these microenterprises would likely accrue to residents of the village.

⁵⁵ Figure A15 plots these sampling distributions, which we convert to 2010 rupees per year. Our NSS expenditure variable subtracts spending on electricity, in order to account for the benefits and costs of electricity consumption. We lack the data to do the same for SHRUG expenditure per capita, meaning that our fuzzy RD estimates may slightly overstate net welfare benefits.

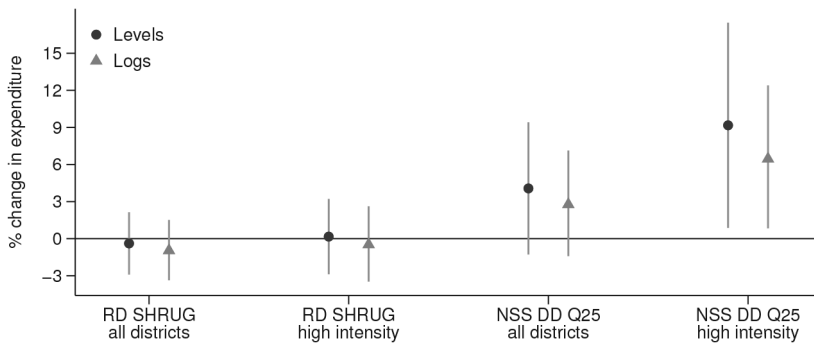


FIG. 8.—Reduced-form expenditure effects, all districts versus high-intensity districts. This figure plots reduced-form estimates for expenditure per capita. For regressions where the outcome variable is in levels (circles), we divide point estimates by within-sample means of the outcome variable to interpret as percent changes. For regressions where the outcome variable is in logs (triangles), we convert point estimates to percent changes (i.e., $\exp(\beta) - 1$). “All districts” estimates pool all 130 RGGVY 10th-Plan districts. “High-intensity” estimates use only the 90 RGGVY 10th-Plan districts where at least 60% of villages received treatment. Whiskers display 95% confidence intervals. We report the corresponding point estimates in (from left to right) tables 4, A15, A24, and A17.

our Q1 and Q25 subsamples) villages. Then, we calculate the present discounted sum of village-wide benefits over 20 years, assuming annual population growth rates from the 2001 and 2011 censuses. We subtract the up-front costs of electrification, using RGGVY’s allowable cost norms: fixed costs of Rs 1.8 million per village, plus Rs 2,200 in variable costs per household (Banerjee et al. 2014).⁵⁶ These numbers are on the low end of rural electrification costs: Lee, Miguel, and Wolfram (2020b) document per-household costs at least an order of magnitude larger in Kenya.

Table 8 reports the share of expenditure draws that generate a positive return on investment (ROI) from rural electrification. For a 300-person village, our fuzzy RD results imply that “full electrification” has less than a 27% chance of generating expenditure benefits that exceed up-front costs, even at with low discount rate of 5%. For a 1,000-person village, our Q1 DD-IV results imply less than a 9% chance of a positive ROI. By contrast, our DD-IV results in the Q25 subsample imply a 90% probability of positive returns from “fully electrifying” a 2,000-person village. This difference is driven primarily by larger per capita treatment effects in Q25 villages.⁵⁷

⁵⁶ Variable costs are based on the allowable costs for below-poverty-line household connections, which were fully subsidized under RGGVY. We inflate these cost norms from 2008 rupees to 2010 rupees.

⁵⁷ Table A25 reports similar results using RGGVY’s “low” fixed-cost norm of Rs 1.3 million per village (Banerjee et al. 2014) and assuming zero population growth.

TABLE 8
RETURN ON INVESTMENT (ROI) FROM ELECTRIFICATION,
USING CONSUMPTION EXPENDITURE

Pr(20-YEAR ROI > 0), BY VILLAGE POPULATION				
	300	300	1,000	2,000
Discount rate:				
$r = .05$	.176	.268	.090	.910
$r = .10$	.159	.240	.089	.909
$r = .15$	.140	.213	.086	.908
Expenditure/capita	SHRUG	SHRUG	NSS	NSS
Endogenous variable	Hr pwr	Brightness	1[HH elec > 0]	1[HH elec > 0]
Instrument	300-person RD	300-person RD	1st-wave district	1st-wave district
Estimation sample	RD BW	RD BW	Quintile 1	Quintiles 2–5

NOTE.—We simulate expenditure benefits using our results from col. 1 of table 6 and cols. 2–3 of table 7. We rescale fuzzy RD estimates by 10 (for hours of commercial power) and 2.6 (for nighttime brightness) and convert all four estimates to annual expenditures per capita in 2010 rupees. Then, we make 10,000 draws from each rescaled sampling distribution (see fig. A15) and calculate the 20-year discounted sum of expenditure changes for villages of 300, 1,000, or 2,000 people. We assume a constant flow of annual benefits in the village, applying annual population growth rates from the 2001 and 2011 censuses. Finally, we subtract up-front fixed and variable costs of electrification. Following Banerjee et al. (2014, 51), we assume fixed costs of Rs 1.8 million per village and variable costs of Rs 2,200 per household, inflating from 2008 to 2010 rupees. Table A25 repeats these simulations under alternative fixed costs and without population growth. Hr pwr = hours of power; HH elec = household electricity consumption; BW = bandwidth.

The gap in expenditure-based ROIs between small and large villages suggests that electrification was more effective at creating new income-generating opportunities in larger communities. We test the extent to which structural transformation explains the ROI gap using village-level outcomes from the Economic Census. Figure 9 plots reduced-form DD coefficients by bins of village population. In smaller villages, we see no evidence that RGGVY drove firm growth. However, in villages larger than 3,000 people, we find 10% increases in the number of firms and 9% increases in the number of firm employees. This suggests that large villages were able to reap the benefits of expanded electricity access by shifting production into firms, whereas this did not occur in smaller villages.

Our second strategy for quantifying the benefits of “full electrification” uses our first-stage DD estimates to model households’ consumer surplus from electricity use. We divide our intensive-margin kilowatt-hour estimates by our extensive-margin estimates for household connections. This scales RGGVY’s impact on household electricity consumption to a “full-electrification” benchmark where 100% of households receive new grid connections, while conservatively assuming that all estimated kilowatt-hour increases come from new household connections. This implies an average monthly consumption per newly connected household of 53.9 kWh in Q1 villages and 73.4 kWh in Q25 villages. Using these kilowatt-hour

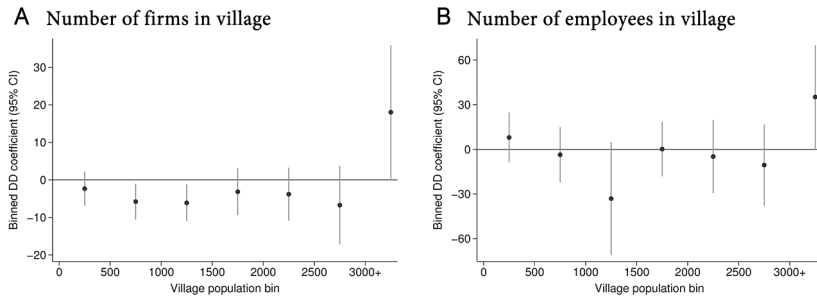


FIG. 9.—Village-level DD of Economic Census outcomes, by village population bin. This figure plots DD coefficients by village population bin for outcomes in panel D of table 4. Regressions use a panel of four Economic Census waves (1990, 1998, 2005, 2013), with 814,715 village-year observations. We interact DD treatment ($1[10\text{th-Plan district}] \times 1[2013]$) with bins of 2001 village population (≤ 500 , 501–1,000, ..., 2,501–3,000, $> 3,000$). In the $> 3,000$ bin, the average village had 187 firms and 408 employees—meaning that our DD estimates for this bin represent 10% and 9% increases, respectively. We interact year fixed effects with population bins and with two sets of quantiles in 2005 expenditure per capita at the district level (within-state quartiles and national deciles). Regressions also include village fixed effects, state-specific linear trends, and year-specific slopes in 2001 village-level covariates (which increase precision). Whiskers display 95% confidence intervals (CIs), with standard errors clustered by district.

quantities, an electricity price of Rs 2.64 per kWh, and a demand elasticity of -0.62 from Burgess et al. (2023), we calculate consumer surplus per household under the assumption of linear demand.⁵⁸

Table 9 presents 20-year internal rates of return (IRRs) from “full electrification,” applying the same cost assumptions as in table 8.⁵⁹ Row 3 reports IRRs for our preferred scenario, which assumes annual growth in population (from the census) and kilowatt-hour consumption. In this scenario, the IRR for a 300-person village is 0%, implying that “full electrification” is welfare reducing under any time discounting. For a 1,000-person village, the 13% IRR is just above the 10%–12% threshold commonly used to benchmark cost-effectiveness (Asian Development Bank 2013; Bonzanigo and Kalra 2014). The 33% IRR for a 2,000-person village far exceeds this threshold, implying large welfare increases from “full electrification.” Our results are similar if we construct kilowatt-hours allowing heterogeneity by expenditures per capita (row 4), allow expenditure growth to translate

⁵⁸ This consumer surplus approach follows Barreca et al. (2016) and Lee, Miguel, and Wolfram (2020b) in applying a Dubin and McFadden (1984) discrete/continuous model. It has the advantage of capturing non-expenditure-related aspects of private utility, such as the ability to read at night using electric lighting.

⁵⁹ Table A26 replicates table 9 under RGGVY’s “low” fixed-cost norm (Banerjee et al. 2014). Table A28 provides more detail on the assumptions entering into these IRR calculations. Table A27 reports first-stage NSS estimates split by expenditure quartiles, used in rows 4–6 of table 9.

TABLE 9
IRRs FROM ELECTRIFICATION, USING CONSUMER SURPLUS

FIRST-STAGE NSS ESTIMATES	SCENARIO	IRRs BY VILLAGE POPULATION (%)		
		300	1,000	2,000
1. Q1 versus Q25 split	No population or kWh growth	...	8	27
2. Q1 versus Q25 splits	No kWh growth	...	10	29
3. Q1 versus Q25 splits	Preferred	0	13	33
4. Expenditure quartile splits	Preferred	1	15	26
5. Expenditure quartile splits	NSS expenditure growth	1	16	28
6. Expenditure quartile splits	3% expenditure growth	2	15	28
7. Q1 versus Q25 splits, districts with high power quality	Preferred	...	13	75

NOTE.—Each row calculates internal rates of return from electrifying all households in villages of 300, 1,000, and 2,000 people. We use econometric estimates to calculate assumed kilowatt-hours per newly electrified household, dividing first-stage kWh/month estimates by first-stage extensive-margin estimates. Rows 1–3 use cols. 2 and 4 from tables A8–A9; rows 4–6 use estimates from table A27 weighted by household expenditure shares for NSS villages of each size; row 7 uses cols. 3–6 of table A20. Using these consumption levels, we apply the methodology of Lee, Miguel, and Wolfram (2020b) to calculate consumer surplus per household. We assume linear demand, a retail electricity price of Rs 2.64 per kWh (the 2010 NSS median price), and a rural electricity demand elasticity of 0.62 (Burgess et al. 2023; Mahadevan [forthcoming] finds a similar rural residential elasticity of 0.56). We calculate IRRs by taking a 20-year discounted sum of consumer surplus in the village, using the same cost assumptions as in table 8 (see table 8 note for details.) Our preferred scenarios apply annual population growth rates from the 2001 and 2011 censuses and 3% annual growth in electricity consumption. In rows 1, 2, and 7, electrification decreases welfare for 300-person villages, even with a 0% discount rate. Table A26 repeats these calculations assuming lower fixed costs per village. Table A28 breaks down the components used to construct these IRRs.

into annual increases in kilowatt-hours consumed (rows 5–6), or restrict our sample to districts with at least 10 hours per day of rural power supply (row 7).⁶⁰ Across a range of assumptions, we find that rural electrification reduces welfare in small villages and yields substantial welfare gains in large villages.

These divergent results for small and large villages help to reconcile conflicting estimates in the existing literature. Lee, Miguel, and Wolfram (2020b) find that electrification is benefit-cost negative in Kenyan villages with an average size of 535 people. In contrast, prior research showing positive impacts of electrification has focused on larger treated populations: approximately 1,200-person villages in the Philippines (Chakravorty, Emerick, and Ravago 2016), similar-sized communities in South

⁶⁰ One might worry that low benefits from electrification are inevitable, given intermittent power supply in Indian villages. Appendix A.3 focuses on districts with at least 10 hours per day of rural power supply in 2011. For these high-power-quality districts, we find larger first-stage impacts on the extensive and intensive margins of household consumption (see table A20). Our DD-IV estimates remain largely unchanged for the Q1 sample. However, they increase in magnitude and gain weak statistical significance for the Q25 subsample (see table A21), suggesting that improved reliability is beneficial in larger villages.

Africa (Dinkelman 2011), entire Brazilian counties (Lipscomb, Mobarak, and Barham 2013), and entire Indian states (Rud 2012).⁶¹

VII. Conclusion

What are the economic effects of expanding electricity access? This paper evaluates the medium-run welfare impacts of electrification in the context of RGGVY, India's flagship national rural electrification program. RGGVY brought electricity access to the world's largest unelectrified population, affecting over 400,000 villages across rural India. The lessons from RGGVY are highly relevant to ongoing electrification efforts, since RGGVY took place while India was at a level of economic development similar to that of the majority of today's unelectrified populations. RGGVY provided connections to the power grid, which likely represents the future of rural electrification despite the emergence of minigrids (Burgess et al. 2023).

Against this backdrop, we demonstrate that RGGVY significantly shrank—though it did not eliminate—India's electricity access gap. Despite increases in electricity access and consumption, the program generated limited economic impacts in the medium term. We scale these program effects using IVs, and we can reject meaningful economic benefits from electrifying small villages.

When considering the full welfare consequences of rural electrification, we find that any benefits that may accrue to small villages do not outweigh the costs of electrifying low-population areas. However, we find that electrification creates sizeable welfare gains in larger villages, likely due in part to structural transformation. These results—from a single massive electrification program—help to reconcile estimates from the literature, which has recently found small economic impacts in villages (e.g., Lee, Miguel, and Wolfram 2020b) and previously found large effects in bigger populations (e.g., Rud 2012).

Our results imply that targeting can help to improve the economic efficiency of last-mile electrification. They signal the potential importance of targeted complementary investments: we find that reliable power supply increases the benefits from electrification in large villages but not in small villages. Our results also speak to other investments besides electricity. While “first-mile” infrastructure projects have been shown to improve welfare in low-income settings (e.g., transportation networks in Donaldson 2018 and Banerjee, Duflo, and Qian 2020), last-mile investments may be less likely to generate meaningful changes in well-being (e.g., rural roads

⁶¹ Summary statistics from Chakravorty, Emerick, and Ravago (2016) suggest an average of 240 households per village, with a mean household size of 5.25, for an approximate village population of 1,200. Dinkelman (2011) reports averages of 630 and 765 adults (aged 15–59) per community. Assuming that households include 5 people on average yields approximately 1,150 people per community.

in Asher and Novosad 2020). Policy makers wishing to provide public goods to rural communities face two challenges: not only are there high costs of providing infrastructure to remote, sparsely populated areas, but these communities may not be able to translate improved infrastructure into meaningful economic gains.

Data Availability

All data and programs required to replicate this paper can be found in Burlig and Preonas (2024) in the Harvard Dataverse, <https://doi.org/10.7910/DVN/APWICK>.

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